Classwork Set #4

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Problem 2

$(x, y, z) \cdot (1, 2, 3) = 0$.

Problem 3

(x, y, z) is in the plane spanned by $u = (3, -1, 2)$ and $v = (-1, 4, 0)$, which means it must be orthogonal to the vector that is orthogonal to both u and v. This vector is given by $u \times v$:

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 2 \\ -1 & 4 & 0 \end{vmatrix} = \hat{i}\,(0 - 8) - \hat{j}\,(0 - (-2)) + \hat{k}\,(12 - 1) = -8\hat{i} - 2\hat{j} + 11\hat{k} = (-8, -2, 11)$$

(x, y, z) is orthogonal to $u \times v = (-8, -2, -11)$ so $(x, y, z) \cdot (-8, -2, -11) = 0$.

Problem 4

We shift the plane by $(1, 2, 3)$, so it passes through that point instead of the origin $(0, 0, 0)$. This means that $(x - 1, y - 2, z - 3) \cdot (-8, -2, -11) = 0$.

Problem 5

I believe such a line does not exist.

The equation of a line passing through the origin and $(3, -1, 2)$ is $(x, y, z) = t(3, -1, 2)$. $(-1, 4, 0)$ is not a scalar multiple of $(3, -1, 2)$, so a line passing through these three points cannot exist.

Problem 6

Line parallel to vector $(-4, 5, -2)$ and passing through $(3, -1, 2)$. Equivalent to, if $t \in \mathbb{R}$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} + t \begin{bmatrix} -4 \\ 5 \\ -2 \end{bmatrix}$$

Problem 7

Set each of $(x - x_0)/a, (y - y_0)/b, (z - z_0)/c$ to the parameter t in a set of parametric equations:

$$\begin{cases} t = \dfrac{x - x_0}{a} \\ t = \dfrac{y - y_0}{b} \\ t = \dfrac{z - z_0}{c} \end{cases} \iff \begin{cases} at = x - x_0 \\ bt = y - y_0 \\ ct = z - z_0 \end{cases} \iff \begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases} \iff \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

This is a line passing through the point (x_0, y_0, z_0), and parallel to the vector (a, b, c).

(I read through the textbook to understand what is going on in questions 6 and 7.)

Problem 8

$2x - 3y + z = 8 \iff 2(x - 4) - 3y + z = 0 \iff (x - 4, y, z) \cdot (2, -3, 1) = 0$. The plane is orthogonal to $n = (2, -3, 1)$.

Let $Q = (Q_x, Q_y, Q_z)$; since Q is on the plane:

$$2Q_x - 3Q_y + Q_z = 8 \iff Q_z = 8 - 2Q_x + 3Q_y$$

From here we have $Q = (Q_x, Q_y, 8 - 2Q_z + 3Q_y)$. We also have vector $QP = P - Q$:

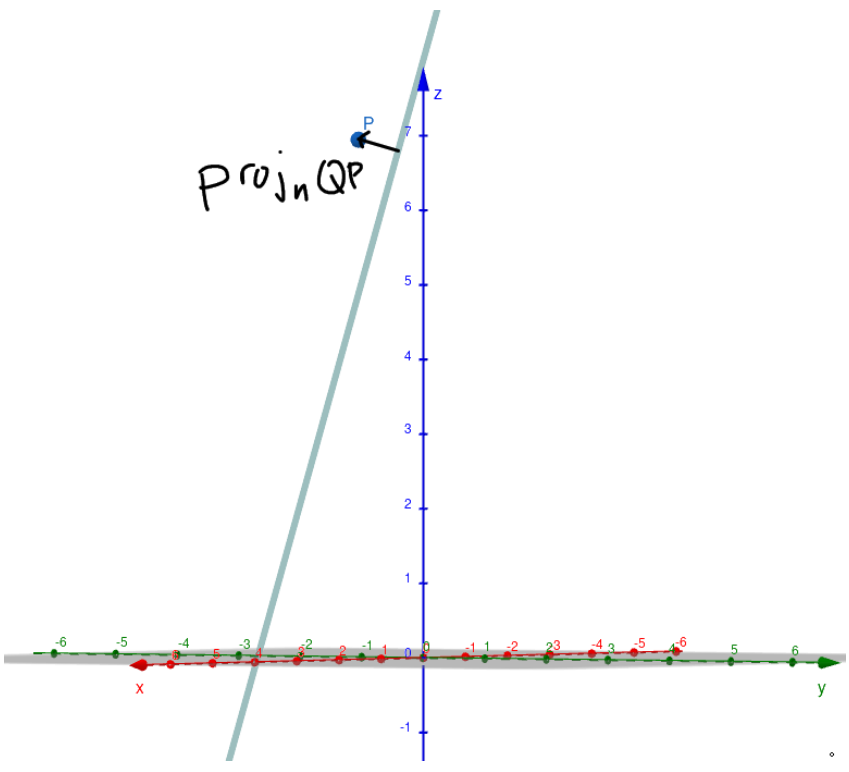
$$QP = \begin{bmatrix} 3 - Q_x \\ 1 - Q_y \\ 7 - (8 - 2Q_x + 3Q_y) \end{bmatrix} = \begin{bmatrix} 3 - Q_x \\ 1 - Q_y \\ -1 + 2Q_x - 3Q_y \end{bmatrix}$$

We can calculate the project $\text{proj}_n QP$:

$$\text{proj}_n QP = \frac{QP \cdot n}{n \cdot n} n = \frac{6 - 2Q_x - 3 + 3Q_y - 1 + 2Q_x - 3Q_y}{2^2 + (-3)^2 + 1^2} n = \frac{2}{14} n = \boxed{\frac{1}{7} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}}$$

Problem 9

$\|\text{proj}_n QP\|$ is the shortest distance between the plane and P.



We can also do this another way to verify this answer. We can find the projection of P onto the plane, and calculate the error. This error should equal $\text{proj}_n QP = (2, -3, 1)/7$.

First, we shift everything for convenience. Instead of projecting P onto the plane, we project $P' = P - (4, 0, 0) = (-1, 1, 7)$ onto the plane formed by $2x - 3y + z = 0$, or:

$$\begin{bmatrix} 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \iff \begin{bmatrix} 1 & -\dfrac{3}{2} & \dfrac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

So this plane is spanned by vectors $(3/2, 1, 0)$ and $(-1/2, 0, 1)$, and a vector v is in the plane if and only if $v \in C(A)$:

$$A = \begin{pmatrix} 3/2 & -1/2 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

We project onto the column space of A, which is the plane. To project the point $P' = (-1, 1, 7)$, we calculate $R = A(A^T A)^{-1} A^T P'$, and the error of the projection is $P' - R$.

Calculator says the error $P' - R = P' - A(A^T A)^{-1} A^T P' = (2, -3, 1)/7$. Great.